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# Speaker-Conditioned Verification and Lexicographic Tie-Breaking for Hallucination Detection in Legislative Hearings

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## Abstract

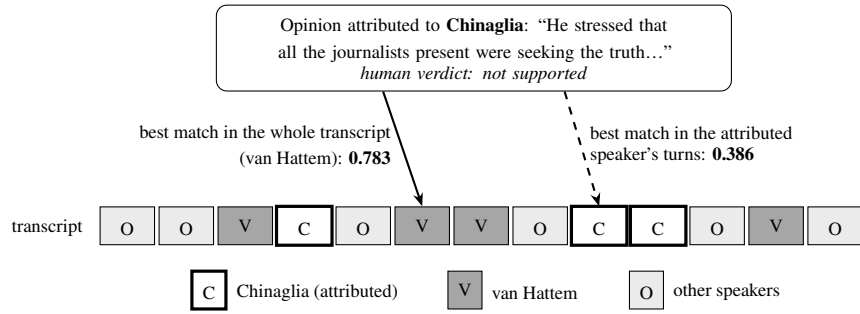
**Abstract.** Summaries of legislative public hearings attribute opinions to named participants, and some of these attributions are not supported by what the participants actually said. We study how to detect such errors on PublicHearingBR, a dataset of Brazilian Chamber of Deputies hearings with human verdicts for every summary opinion. We first show that standard verification scores, which compare an opinion with the whole transcript and keep the best match, cannot distinguish an opinion expressed by the attributed speaker from the same opinion expressed by someone else. Features that compare the opinion with the attributed speaker’s own turns raise the AUROC (the probability of ranking a hallucination above a supported opinion) from 0.57 to 0.76 without any model call. We then study how to add such an inexpensive detector to a committee of twelve LLM judges. Fitting a combined model degrades the committee, because a weighted sum can overturn its decisions. Using the inexpensive score only to order opinions on which the committee is tied cannot do so, and an elementary decomposition of AUROC forecasts the gain of this rule on unseen hearings. The rule improves every committee size we tested, and its gain is largest where judges are fewest: it raises each of the twelve judges individually by 0.046 to 0.102 AUROC (a typical judge from 0.81 to 0.88), and in an exploratory analysis seven judges with the rule are statistically indistinguishable from the full committee of twelve. On this dataset, a small committee with the rule therefore approaches the accuracy of a much larger one. The rule also raises the full committee from 0.926 to 0.930, a small gain that is stable across twenty random assignments of hearings to folds.

**Keywords:** LLM judges; hallucination detection; speaker incidence; AUROC ties; lexicographic combination; PublicHearingBR

## 1 Introduction

Legislative public hearings are long events with many speakers. When they are summarised, whether by journalists or by language models, each summary sentence usually attributes an opinion to a named participant. An opinion that the participant did not express is a hallucination with public consequences: it attributes words to an identifiable person. Detecting such errors requires comparing each attributed opinion with what that person actually said.

PublicHearingBR [5] makes this problem measurable. It contains 206 transcripts of public hearings of the Brazilian Chamber of Deputies, a structured summary of each hearing, and a human verdict for each of the 4,238 summary opinions: 504 of them (11.9%) are marked as not supported by the transcript. The dataset also provides the verdicts of twelve LLM judges (three prompts applied to four models); each judge is a language model asked whether the opinion is supported, and it returns a binary vote. We



**Figure 1.** A real misattribution from hearing data\_001. The order of turns in the strip is schematic; the two similarity scores are measured. A score computed over the whole transcript finds strong support (0.783), but that support comes from another deputy. Restricted to the attributed speaker’s turns, the best match drops to 0.386.

measure detection quality by the AUROC, the probability that a randomly chosen hallucination is scored above a randomly chosen supported opinion (0.5 is chance and 1 is perfect; Section 3.1). A single judge reaches an AUROC between 0.734 and 0.847, and the sum of the twelve votes, which we compute from the released verdicts, reaches 0.926, the strongest result we are aware of on this task. Unless stated otherwise, every AUROC in this paper is computed on the 3,630 opinions whose speaker could be identified in the transcript (Section 5). Each judge call has a cost, however, and it is not known how inexpensive signals should be used alongside the judges.

Inexpensive detectors compare the opinion with each sentence of the transcript, using embedding similarity or natural language inference (NLI), and keep the best match. On this task they reach an AUROC of 0.57 (similarity) and 0.58 (entailment) (Section 6), far below the judges. The reason is that a plausible hallucination is on topic: the hearing did discuss the subject, so some sentence of the transcript resembles the opinion.

Our starting observation is that the best match may belong to the wrong person. Figure 1 shows a real case. In a hearing on press freedom, the summary attributes to deputy Arlindo Chinaglia the opinion “*Destacou que todos os jornalistas presentes estavam em busca da verdade, independentemente de suas preferências políticas*” (“He stressed that all the journalists present were seeking the truth, regardless of their political preferences”), and the human annotator marked it as not supported. The most similar sentence in the transcript has a cosine similarity of 0.783, but it was spoken by deputy Marcel van Hattem. The most similar sentence among Chinaglia’s own turns scores 0.386 and concerns a different point. A score computed over the whole transcript sees a well-supported opinion; a score computed over the attributed speaker’s turns sees an unsupported one.

This is not a property of one example. Any score that combines sentence-level scores through a maximum, a mean or a similar function ignores which speaker produced each sentence, and therefore cannot separate the two situations (Observation 6). The remedy is to condition on the speaker: we compare the opinion with the attributed speaker’s turns and with the other speakers’ turns, and use these comparisons as features. We call them *speaker-incidence features*, because they use which speaker uttered each sentence. The best match in the attributed speaker’s turns alone raises the AUROC from 0.57 to 0.75, and a logistic regression on all the features reaches 0.76, without any model call.

The second question is how to combine such a detector with the committee of judges. The usual approach fits a model, such as a logistic regression, on the judges’ votes and the new features together. On this dataset it makes the committee worse. The reason is structural. The committee’s vote count takes only thirteen values, so many opinions are tied, and a second detector could usefully order them. A weighted sum, however, does more than order the ties: it can also reverse two opinions that the committee orders correctly, whenever the new detector disagrees strongly enough (Proposition 7). We instead rank opinions by the committee’s vote count and use the new detector only to order opinions that received the same count. This *lexicographic* combination never reverses a committee decision, and its AUROC is given by an exact formula (Proposition 10) that depends on two quantities measurable without running the combination: how often the committee cannot order a pair of opinions, and how well the new detector orders those pairs.

### Contributions.

- Speaker-incidence features, motivated by the observation that pooled verification scores ignore speaker attribution, which raise AUROC by 0.18 at no model cost (Sections 4.1 and 6).

- A rule for combining a reliable but coarse detector with an inexpensive but finer one. The rule never overturns the reliable detector, and an elementary decomposition of AUROC gives its gain exactly (Section 4.2). Measured on one half of the hearings, the decomposition forecasts the gain on the other half without bias, at every committee size (Section 7.4).
- A large reduction in the cost of accurate detection. The rule improves every committee size we tested: it raises each single judge by 0.046 to 0.102 AUROC, and in an exploratory analysis seven judges with the rule are statistically indistinguishable from the full committee of twelve (Section 7.4).
- A characterisation of what remains: on the pairs of opinions that the committee orders incorrectly, the inexpensive detector performs below chance, so rules that overrule the committee when the two disagree cannot recover them (Section 7.5).
- An evaluation in which predictions were committed to the repository before each experiment and failed predictions are reported alongside successful ones (Section 8), with all code and prediction files publicly available.<sup>1</sup>

The paper is organised as follows. Section 2 reviews related work. Section 3 introduces the evaluation metric and the notation. Section 4 states the theoretical results. Section 5 describes the data and the experimental protocol. Sections 6 and 7 report the results for the speaker-incidence features and for their combination with the judges. Sections 8 and 9 report negative results and threats to validity, and Section 10 concludes.

## 2 Related Work

**Hallucination detection and LLM judges.** Fernandes et al. [5] introduce the corpus, the human verdicts and the twelve LLM-judge annotations used here. Maynez et al. [9] propose the standard taxonomy of faithfulness errors in abstractive summarisation. The error our features target, content attributed to the wrong speaker, is identified in dialogue summarisation by Ramnath et al. [11], who report that LLM judges detect it only moderately well. Zheng et al. [14] study the reliability of LLM judges, including position, verbosity and self-preference biases. Our work differs in two respects. We condition verification on the speaker to whom an opinion is attributed, which, to our knowledge, has not been evaluated on this task. We also treat the judges’ categorical verdicts as a coarse ranking whose ties a second detector can resolve (Section 4.2).

**Combining detectors.** The question of when a classifier should abstain goes back to Chow [2], with the risk–coverage formalism of El-Yaniv and Wiener [4]. A coarse detector abstains implicitly when it assigns the same score to two items, and Lemma 2 quantifies the effect of these ties on AUROC. Ensemble and rank-aggregation methods usually combine scores by weighted sums. The decomposition of AUROC into tied and separated pairs that we use is elementary; its value is practical, since it indicates where ordering the ties of a coarse detector can help. Noh et al. [10] compare scoring the claims of a summary jointly or independently and find the two comparable; they vary what a scorer reads, whereas we vary how two scorers are combined.

**Topological data analysis.** Topological methods have been applied to hallucination detection through model internals, such as attention graphs [1] and layer-wise persistence [13]. We use only model outputs and the transcript. We also tested structural features computed from the relation between opinions and speakers, including its homology [3] and a Hodge decomposition [7]; they did not improve the results (Section 8). The proofs in this paper use only the rankings induced by the scores, and no property of the data or of the models, so they hold for any primary and secondary detector and can be checked on a new deployment by counting pairs.

## 3 Background

This section introduces the evaluation metric and the notation used in the rest of the paper. The definitions are standard. Lemma 2 restates the identity of Hanley and McNeil [6]; it is included so that the later results can be checked, and it is not claimed as a contribution.

<sup>1</sup><https://github.com/caua-sathler/IdeiasEmRede-TopoIA>

### 3.1 AUROC as a statement about pairs

A *gold opinion* is a sentence of a structured summary attributed to a named participant, such as the opinion attributed to deputy Chinaglia in Figure 1. A human annotator marked each opinion as supported by the transcript or not. We call the unsupported opinions *hallucinations* (positives) and the supported ones *grounded* (negatives). A *detector* is any function that assigns a real number to each opinion, with higher values meaning more suspect. A single binary verdict and a committee’s vote count are also detectors; they simply take few distinct values.

**Definition 1** (AUROC). Let  $P$  be the set of positive items,  $N$  the set of negative items,  $X = P \cup N$ , and  $s : X \rightarrow \mathbb{R}$  a score in which larger values indicate a positive. Following Hanley and McNeil [6], who identify the area under the ROC curve with the Mann–Whitney statistic,

$$\text{AUROC}(s) = \Pr[s(p) > s(n)] + \frac{1}{2} \Pr[s(p) = s(n)], \quad (1)$$

where  $p$  and  $n$  are drawn independently and uniformly from  $P$  and  $N$ .

In words, AUROC is the probability that a randomly chosen hallucination receives a higher score than a randomly chosen grounded opinion, with ties counted as one half. A value of 0.5 corresponds to chance, 1.0 to a perfect ranking, and a value below 0.5 to a ranking in the wrong direction. We use AUROC rather than accuracy because positives are rare: a detector that flags nothing is 88.1% accurate yet detects nothing. AUROC is also unchanged by any strictly increasing transformation of the score, so detectors do not need to be calibrated. Two properties are used throughout. First, AUROC is an average over positive–negative pairs, so it can be decomposed over any partition of those pairs (Section 7.2). Second, the **tie fraction**  $\tau(s) = \Pr[s(p)=s(n)]$  contributes exactly  $\tau/2$ .

**Lemma 2** (Tie accounting). Let  $A(s) = \Pr[s(p) > s(n)]$ . Then  $\text{AUROC}(s) = A(s) + \tau(s)/2$ . Hence  $\text{AUROC}(s) \leq 1 - \tau(s)/2$ , with equality if and only if  $s$  correctly orders every pair it separates, that is, every pair to which it assigns different scores.

*Proof.* The identity is (1). Separated pairs have total probability  $1 - \tau$ , so  $A \leq 1 - \tau$ ; substituting into the identity gives the bound, with equality exactly when  $A = 1 - \tau$ , that is, when every separated pair is ordered correctly.  $\square$

**Remark 3.** For a binary detector, AUROC equals the balanced accuracy,  $\frac{1}{2}(\text{sensitivity} + \text{specificity})$ , so the bound  $1 - \tau/2$  says nothing about how close a judge is to a perfect detector. For example, a judge that flags half of the hallucinations and never flags a grounded opinion has sensitivity 0.5, specificity 1 and AUROC 0.75. It ties every pair formed by a missed hallucination and a grounded opinion, so  $\tau = 0.5$  and the bound is  $1 - \tau/2 = 0.75$ : the judge attains the bound exactly, yet misses half of the hallucinations. What the lemma provides for any detector, binary or not, is that  $\tau/2$  is the largest gain obtainable by ordering its tied pairs while leaving the separated pairs unchanged (Section 7).

### 3.2 Natural language inference

A **natural language inference** (NLI) model reads an ordered pair of sentences (premise, hypothesis) and returns three probabilities:

| label         | meaning                               | example (premise $\rightarrow$ hypothesis)           |
|---------------|---------------------------------------|------------------------------------------------------|
| entailment    | the premise makes the hypothesis true | “it was in 1988” $\rightarrow$ “it was in the 1980s” |
| contradiction | they cannot both be true              | “it was in 1988” $\rightarrow$ “it was in 1995”      |
| neutral       | neither of the above                  | “it was in 1988” $\rightarrow$ “it rained that day”  |

Entailment is directed: “1988” entails “the 1980s”, but not conversely. Contradiction is symmetric. We use the entailment probability as a feature (Section 5). Each evaluation of a sentence pair is one *NLI call*.

### 3.3 Pooled scores and rankings with ties

Most inexpensive verification methods compute a sentence-level score  $f(o, u_i)$  between the opinion  $o$  and each transcript sentence  $u_i$ , such as a cosine similarity or an entailment probability, and then *pool* these scores with a maximum, a mean, or the mean of the  $k$  largest values. These pooling functions are **permutation-invariant**: their value depends only on the collection of sentence scores, not on which

sentence produced which score. Any information that the collection does not record, such as the speaker of each sentence, is therefore invisible to them. Section 4.1 makes this precise.

The combination argument of Section 4.2 requires one more notion. Informally, a *preorder* is a ranking that allows ties: a committee’s vote count, for instance, ranks opinions but assigns the same value to many of them.

**Definition 4** (Preorder). A relation  $\leq$  on a set is a **preorder** if it is reflexive and transitive. Distinct items  $x, y$  may satisfy  $x \leq y \leq x$ , written  $x \approx y$ : they are *tied*. We write  $x < y$  when  $x \leq y$  and not  $y \leq x$  (a *strict* relation). The groups of tied items are the **equivalence classes** of the preorder. A preorder is **total** if every two items are comparable, and it **refines** another preorder if it keeps all the strict relations of the other and only splits some of its ties. (These two terms are used in Proposition 10.)

A score  $s$  induces the **detector preorder**  $x \leq_s y \iff s(x) \leq s(y)$ . The results below depend only on this preorder, so they are unchanged by any strictly increasing transformation of  $s$ : a binary verdict and a continuous probability are the same kind of object and differ only in the number of equivalence classes.

**Example 5** (A committee is coarse). The sum of twelve binary votes takes thirteen values. Of the 3,630 opinions evaluated in Section 5, 52.3% receive zero votes, so half of the dataset lies in a single equivalence class, within which the committee expresses no preference. This does not imply a large tie fraction  $\tau$ , because  $\tau$  counts positive–negative pairs that share a class, not items that share a class; pairs of two grounded opinions do not enter AUROC. Only 19 of the 408 positives receive zero votes, against 1,878 negatives, so this class contributes  $19 \times 1,878 = 35,682$  tied pairs, about 2.7% of all  $408 \times 3,222$  positive–negative pairs. Over all thirteen classes,  $\tau = 4.65\%$  (Section 7, Figure 2).

**Notation.** The symbols used in the rest of the paper are summarised below.

|                  |                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------|
| $P, N$           | positive (hallucinated) and negative (grounded) opinions                                                  |
| $s; s_1, s_2$    | a detector’s score; the primary detector (the judges’ vote count) and the secondary, inexpensive detector |
| $\tau$           | tie fraction: share of positive–negative pairs that receive equal scores                                  |
| $\alpha$         | AUROC of $s_2$ restricted to the pairs that $s_1$ ties                                                    |
| $T, u_i, T_a$    | a transcript, its sentences, and the sentences spoken by participant $a$                                  |
| $\sigma$         | speaker map, assigning each sentence to its speaker                                                       |
| $\cos_s, \Delta$ | best match in the attributed speaker’s turns; misattribution score (Section 5)                            |
| $k$              | number of judges in a committee                                                                           |

## 4 Theory

### 4.1 Pooled scores ignore the speaker

Fix an opinion  $o$  attributed to speaker  $a$ , a transcript  $T = (u_1, \dots, u_m)$  with a speaker map  $\sigma : T \rightarrow \mathcal{A}$  that assigns each sentence to the participant who uttered it, and any sentence-level score  $f(o, u_i)$ .

**Observation 6** (Incidence blindness). *Any pooled score  $G(\{f(o, u_i)\}_i)$ , where  $G$  is a maximum, a mean, a top- $k$  mean or any other permutation-invariant function, is invariant under every reassignment of the speaker map  $\sigma$ . Consequently, no such score can distinguish an opinion supported by the turns of its attributed speaker from the same opinion supported only by the turns of other speakers.*

*Proof.*  $G$  does not receive  $\sigma$  as an argument. For a concrete separation, take two sentences with scores 0.9 and 0.1. Under  $\sigma_1$  the first sentence belongs to  $a$ ; under  $\sigma_2$  it belongs to some  $b \neq a$ . Every pooled  $G$  takes the same value on both configurations, but only the first one supports the opinion as attributed.  $\square$

Equivalently, exchanging the speaker labels of all turns in a hearing leaves every pooled score unchanged, although it changes the answer to the question of whether the attributed person said what the opinion claims. No choice of pooling function avoids this. The remedy is to pool over the attributed speaker’s sentences  $T_a = \sigma^{-1}(a)$  instead of the whole transcript, and to compare the result with the other speakers’ sentences. Section 6 shows that this restriction raises AUROC by 0.18, and that additional features describing how the support is distributed across speakers add a smaller, significant improvement.

## 4.2 Combining a coarse and a fine detector

Consider a *reliable but coarse* detector  $s_1$ , such as the committee, which is accurate when it separates two items but assigns the same value to many of them, and an *inexpensive but fine* detector  $s_2$ , which separates almost all items but is less accurate. We call  $s_1$  the **primary** and  $s_2$  the **secondary** detector throughout. The standard way to combine them is a weighted sum, or a logistic regression on both scores, which we call *stacking*.

**Proposition 7** (A weighted sum can reverse the primary detector). *Let  $w_1, w_2 > 0$ . A pair with  $x <_{s_1} y$  is reversed by the weighted sum,  $w_1 s_1(x) + w_2 s_2(x) > w_1 s_1(y) + w_2 s_2(y)$ , if and only if  $s_2(x) - s_2(y) > (w_1/w_2)(s_1(y) - s_1(x))$ . Whenever  $s_2$  is not a non-decreasing function of  $s_1$ , which is the only case in which combining the two can add information, such pairs exist once  $w_2/w_1$  is large enough.*

**Example 8** (A concrete reversal). Consider three opinions. Let  $s_1$  be the committee's vote count divided by twelve and  $s_2$  an inexpensive score in  $[0, 1]$ , with equal weights.

|                         | $x$       | $y$       | $z$       |                           |
|-------------------------|-----------|-----------|-----------|---------------------------|
| committee $s_1$         | 4/12      | 5/12      | 5/12      | $x <_1 y \approx_1 z$     |
| inexpensive score $s_2$ | 0.95      | 0.10      | 0.90      |                           |
| sum $s_1 + s_2$         | 1.28      | 0.52      | 1.32      | <b>reverses</b> $x <_1 y$ |
| lexicographic           | (4, 0.95) | (5, 0.10) | (5, 0.90) | keeps $x < y < z$         |

The sum ranks  $x$  (four votes) above  $y$  (five votes) because the inexpensive score disagrees with the committee. The lexicographic order keeps the committee's decision and uses  $s_2$  only to separate  $y$  from  $z$ , which the committee could not order.

**Definition 9** (Lexicographic combination). Given detector preorders  $\leq_1$  and  $\leq_2$ ,

$$x \leq_{\text{lex}} y \iff x <_1 y \text{ or } (x \approx_1 y \text{ and } x \leq_2 y). \quad (2)$$

Operationally: rank items by  $s_1$ , and within each group of items with equal  $s_1$ , rank them by  $s_2$ .

**Proposition 10** (Lexicographic combination).  $\leq_{\text{lex}}$  is a total preorder containing  $\leq_1 \cap \leq_2$ , and

- (i) it never reverses a strict relation of  $\leq_1$ ;
- (ii) it refines  $\leq_1$  only within its equivalence classes, and uses no other information;
- (iii) if  $\alpha$  denotes the AUROC of  $s_2$  restricted to the positive–negative pairs that  $\leq_1$  ties, then

$$\text{AUROC}(\leq_{\text{lex}}) = \text{AUROC}(s_1) + \tau(s_1)(\alpha - \frac{1}{2}) \quad \text{exactly.} \quad (3)$$

The combination therefore recovers the fraction  $2\alpha - 1$  of the maximum possible gain  $\tau/2$ . It attains the full  $\tau/2$  if and only if  $\alpha = 1$ , and in all cases  $\text{AUROC}(\leq_{\text{lex}}) \in [\text{AUROC}(s_1) - \tau/2, \text{AUROC}(s_1) + \tau/2]$ .

In words, the gain equals the share of tied pairs times how much better than a coin flip the secondary detector orders them.

*Proof.* Totality and transitivity of (2) follow directly from the definition. For the inclusion, let  $x \leq_1 y$  and  $x \leq_2 y$ : either  $x <_1 y$  and the first clause applies, or  $x \approx_1 y$  and the second applies. Claim (i) is the first clause, and claim (ii) the second. For (iii), let  $A = A(s_1)$ , so that  $\text{AUROC}(s_1) = A + \tau/2$  by Lemma 2. By (i), the pairs that  $s_1$  separates are ordered in the same way by  $\leq_{\text{lex}}$  and contribute  $A$ . By (ii), the tied pairs, of total probability  $\tau$ , are ordered by  $s_2$  alone; applying Definition 1 to this sub-population, they contribute  $\tau\alpha$  instead of  $\tau/2$ . Hence  $\text{AUROC}(\leq_{\text{lex}}) = A + \tau\alpha$ , and subtracting  $\text{AUROC}(s_1)$  gives (3). The bound follows from  $\alpha \in [0, 1]$ .  $\square$

**Remark 11** (The gain depends on  $\alpha$ , not on the global AUROC of  $s_2$ ). The gap  $1 - \tau/2 - \text{AUROC}(s_1)$  comes from pairs that  $s_1$  separates in the wrong order, and by (i) the lexicographic combination cannot change them. The largest possible gain is therefore  $\tau/2$ , whatever the secondary detector. Equation (3) shows that, within this bound, the gain is determined by a single number,  $\alpha$ , and that the global AUROC of  $s_2$  does not enter it. A candidate secondary detector should therefore be evaluated on the primary detector's ties rather than on the whole task. The effect is symmetric: a value of  $\alpha$  below  $\frac{1}{2}$  loses exactly what the same distance above  $\frac{1}{2}$  gains. Finally,  $\tau$  depends only on the primary detector, so the maximum gain  $\tau/2$

can be computed before any secondary detector is built; once  $\alpha$  is measured on the ties, Equation (3) gives the gain without computing the combination. Section 7.4 tests this as a forecast on unseen hearings.

**Remark 12** (A family of combinations between the two extremes). Let  $r_1$  be an item’s equivalence class under  $\leq_1$ , numbered by consecutive integers from lowest to highest, and let  $r_2 \in [0, 1]$  be the rank of  $s_2$  divided by the number of items. Combine them as  $f_\gamma = r_1 + \gamma r_2$ . For  $0 < \gamma < 1$ ,  $r_1(x) < r_1(y)$  implies  $f_\gamma(x) < f_\gamma(y)$ , since the classes differ by at least one and  $\gamma |r_2(x) - r_2(y)| < 1$ ; within a class,  $f_\gamma$  orders by  $s_2$ . Hence  $f_\gamma$  coincides with  $\leq_{\text{lex}}$ . (At  $\gamma = 1$ , two items in adjacent classes with  $r_2 = 1$  and  $r_2 = 0$  would tie.) As  $\gamma \rightarrow \infty$ ,  $f_\gamma$  approaches the weighted sum of Proposition 7. The parameter  $\gamma$  is thus the number of classes of  $\leq_1$  by which the secondary score may move an item. For any  $\gamma$ , a pair separated by more than  $\gamma$  classes can be neither reversed nor tied, so  $|\text{AUROC}(f_\gamma) - \text{AUROC}(s_1)| \leq \tau/2 + \beta_\gamma$  with  $\beta_\gamma = \Pr[0 < g \leq \gamma]$ , where  $g$  is the number of classes separating a positive–negative pair. Like  $\tau$ , this bound can be computed from  $s_1$  alone.

## 5 Data and Methods

**Data.** We use all 206 hearings of PublicHearingBR: 4,238 opinions with a human verdict (504 positive, 11.9%) and the stored votes of the twelve LLM judges. Sentences and opinions are embedded with multilingual MPNet [12]. A *turn* is a block of consecutive sentences spoken by one participant. Turns are recovered from the transcript’s speaker markers (O SR. NAME --): each sentence is assigned to the most recent marker. The participant named in the summary is matched to a turn speaker by overlap of name tokens. The match succeeds for 85.7% of the opinions ( $n = 3,630$ , 408 positive), which form the main evaluation set, since both the speaker-incidence features and the combination experiments require the attributed speaker’s turns. A failed match is itself informative and is used as a feature in Section 6. Entailment probabilities come from mDeBERTa-v3-base-xnli [8].

**Statistical protocol.** All learned models are evaluated *out of fold*: we use five-fold cross-validation in which all opinions of a hearing fall in the same fold (GroupKFold), so each opinion is scored by a model fitted on other hearings and no model is evaluated on data it was fitted on. Models are logistic regressions with balanced class weights. Confidence intervals resample whole hearings rather than opinions, since opinions from the same transcript are not independent (3,000 to 5,000 bootstrap resamples, depending on the analysis; all intervals are 95% and, when two detectors are compared, paired on the same resamples). Two additional sources of variance are controlled in the combination experiments. A random tie-break is itself a random variable, so we compare against 200 random draws. The assignment of hearings to folds is not captured by the bootstrap, so every combination result is repeated over 20 random assignments of hearings to folds (*fold partitions*) and reported as median and range. Where a single value is reported, it comes from one representative partition. Scripts that use different fold partitions report slightly different values for the same quantity, for example 0.7605 and 0.7617 for the speaker-incidence features, or  $\alpha = 0.5856$  on the representative partition against a median of 0.589 over twenty partitions; such differences, in the third decimal place, are fold noise and not different quantities. The predictions for each experiment were committed to the repository before it was run (we call them *pre-specified*); those that failed are reported in Section 8.

**Speaker-incidence features.** For an opinion  $o$  attributed to the matched speaker  $a$ , let  $S(o, \cdot)$  be the cosine similarity between  $o$  and each transcript sentence,  $T$  the set of all sentences, and  $T_a$  the set of sentences spoken by  $a$ .

|                              |                                                                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------|
| $\text{cos}_g$               | $\max_{s \in T} S(o, s)$ : best match in the whole transcript (baseline)                                         |
| $\text{cos}_s$               | $\max_{s \in T_a} S(o, s)$ : best match in the attributed speaker’s turns                                        |
| $\Delta$                     | $\max_{b \neq a} \max_{s \in T_b} S(o, s) - \text{cos}_s$ : how much better another speaker supports the opinion |
| $\text{share}_a$             | sum of the similarities of the ten best global matches that lie in $T_a$ , divided by their total                |
| $\text{rank}_a$              | rank of $a$ among all speakers ordered by their best match, scaled to $[0, 1]$                                   |
| $\text{cover}_a$             | 1 if $a$ is among the strong supporters, i.e. the speakers with at least one sentence of similarity $\geq 0.65$  |
| $\text{nli}_s, \text{nli}_g$ | maximum entailment probability over the five best-matching sentences of $a$ / of the whole transcript            |

The feature  $\text{cos}_s$  asks whether the attributed speaker’s own turns support the opinion, and  $\Delta$  asks whether somebody else supports it better; we call  $\Delta$  the *misattribution score*. When a feature is used alone

as a detector, it is oriented so that larger values indicate a more suspect opinion (for example,  $-\cos_s$ ). The remaining features describe how much of the support belongs to the attributed speaker, where that speaker ranks, and whether the speaker is among the supporters at all. None of these quantities can be computed from a pooled score (Observation 6). The cosine features require no model call beyond the precomputed embeddings; the two entailment features require ten NLI calls per opinion.

**The two detectors combined in Section 7.** The primary detector  $s_1 \in \{0, \dots, 12\}$  is the number of judges that flag an opinion. This unweighted vote count outperforms logistic weights fitted out of fold on the same votes (0.9260 vs. 0.9245 on the representative partition, and a median of 0.9226 over twenty partitions, Table 3). The secondary detector is a logistic regression on all the speaker-incidence features, including the two NLI features (0.7792, ten NLI calls per opinion); unless stated otherwise, Section 7 uses this detector. *Stacking* denotes a logistic regression fitted out of fold on the votes and the same features together. As a control, Section 7.7 also uses a deliberately weak inexpensive detector,  $\cos_g$ .

## 6 Results I: Speaker-Incidence Features

### 6.1 Conditioning on the attributed speaker

Table 1 reports the main result of this section. On the 3,630 matched opinions, the best match in the whole transcript reaches an AUROC of 0.5676 (0.5628 over all 4,238 opinions). Computing the same similarity over the attributed speaker’s turns raises it to 0.7455, a difference of +0.178 (clustered CI [+0.128, +0.237]).

**Table 1.** AUROC of each signal against the human verdict on the matched-speaker subset ( $n = 3,630$ , 408 positives). “0 calls” means the signal is computed from precomputed embeddings and speaker markers alone.

| Signal                                                         | Cost / opinion | AUROC         |
|----------------------------------------------------------------|----------------|---------------|
| $\cos_g$ : best match in the whole transcript (baseline)       | 0 calls        | 0.5676        |
| $nli_g$ : best entailment in the whole transcript (baseline)   | ~10 NLI        | 0.5828        |
| $\cos_s$ : <b>best match in the attributed speaker’s turns</b> | 0 calls        | <b>0.7455</b> |
| $\Delta$ : misattribution score                                | 0 calls        | 0.7595        |
| $share_a$ : share of the support                               | 0 calls        | 0.7135        |
| $rank_a$ : rank of the speaker                                 | 0 calls        | 0.7258        |
| $cover_a$ : speaker among strong supporters                    | 0 calls        | 0.6706        |
| <b>all speaker-incidence features, out of fold</b>             | 0 calls        | <b>0.7605</b> |
| all features + $nli_s$ , $nli_g$                               | ~10 NLI        | <b>0.7792</b> |

Three controls address alternative explanations. **Speaker identity.** Computing the same feature over a *randomly chosen other* speaker of the same hearing, matched for number of turns, gives 0.5209, which is at chance and below the global baseline. The paired difference between the attributed and the random speaker is +0.2246 [+0.1903, +0.2592], so the signal comes from *who* spoke, not from how much. **Number of sentences.** The number of sentences of the attributed speaker alone scores 0.4953 (correlation +0.26 with  $\cos_s$ ), so the effect is not an artefact of taking a maximum over fewer sentences. **Features beyond the single score.** Out of fold, the full set of features improves on the best single conditioned score, from 0.7447 to 0.7605, a paired difference of +0.0159 [+0.0028, +0.0298].

Two further observations. First, a failed speaker match is informative (hallucination rate 15.8% vs. 11.2%), and using it as a feature extends the detector to all 4,238 opinions (0.7435 out of fold). Second, conditioning also helps entailment:  $nli_s$  improves on  $nli_g$  by +0.1264 [+0.0928, +0.1614]. Contrary to our pre-specified prediction, however,  $nli_s$  alone (0.7095) does not outperform  $\cos_s$  (0.7455); entailment adds value only in combination with the cosine features (+0.0186 [+0.0060, +0.0309]).

### 6.2 Comparison and combination with individual judges

The twelve judges each emit a binary vote and reach an AUROC between 0.7340 and 0.8468. Without any model call, the speaker-incidence features (0.7605) outperform the three weakest judges (0.7340, 0.7401, 0.7522). Combined with any single judge by *rank sum*, which adds the two ranks and requires no fitting, they improve that judge in twelve cases out of twelve, with every clustered CI excluding zero. The gains range from +0.027 to +0.094, are largest for the weakest judges (per-judge results are in the repository), and the best such rank-sum combination reaches 0.8823. Section 7.4 reports the lexicographic combination, which performs better. Section 7 examines the full twelve-judge committee and how to combine the features with it without degrading it.

**Interpretation.** One type of error these features detect has a simple description: a plausible hallucination that is on topic for the hearing but not for the turns of the person to whom it is attributed. A global similarity score cannot see this difference (Observation 6), whereas the speaker-conditioned features are built to capture it. That the strongest single feature is the misattribution score  $\Delta$  (0.7595) is consistent with this interpretation. The example of Figure 1 (0.783 in the whole transcript, 0.386 in the attributed speaker’s turns,  $\Delta = +0.40$ ) is one such case, and the features flag it, in agreement with the human verdict. Section 6.3 measures how common this type of error is among the positives.

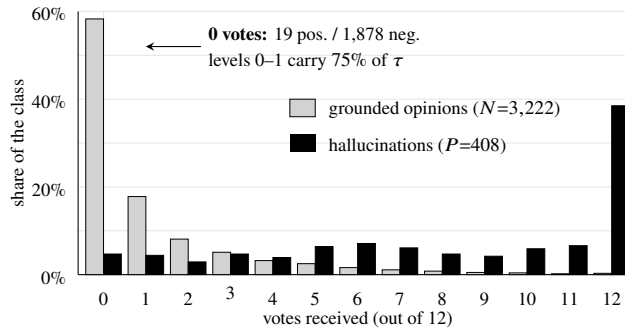
### 6.3 What kind of error are the positives?

To estimate how common misattribution is, we classified a random sample of 50 of the 408 positives, using two independent LLM-assisted passes that agree on 45 items (Cohen’s  $\kappa = 0.81$ ).<sup>2</sup> About a quarter of the positives (12 of 50) are misattributions: the content was expressed in the hearing, but by another participant. On these items  $\Delta$  is always positive (mean 0.21, against 0.05 on the other items), as the interpretation above predicts. The remaining positives are mostly cases in which the attributed speaker discussed the content, and whose label reflects the standard of the original annotation, which we use as given (Section 9).

## 7 Results II: Combining the Features with the Committee

### 7.1 How the committee’s ties are distributed

The sum of the twelve votes takes thirteen values and reaches an AUROC of 0.9260. The ranking it induces is nevertheless coarse:  $\tau = 4.65\%$  of positive–negative pairs are tied, so by Lemma 2 the largest possible gain from ordering them is  $\tau/2 = 0.0232$ , and a perfect secondary detector would reach 0.9492 (not the bound  $1 - \tau/2 = 0.9768$  of Lemma 2, which would also require the committee to order all separated pairs correctly). This does not mean that the judges are close to a performance ceiling (Remark 3): their sensitivity ranges from 0.48 to 0.92 and their specificity from 0.64 to 0.99, and several of them miss more than half of the hallucinations. All gains reported below are bounded by  $\tau/2$ , not by that larger gap.



**Figure 2.** The committee’s tie structure. For each of the thirteen vote levels, the share of grounded opinions (grey) and of hallucinations (black) receiving that many votes. The tie fraction is  $\tau = \sum_k \pi_k^+ \pi_k^-$ , the sum of products of the two bars at each level: level 0 alone contributes 58% of it ( $\pi_0^- = 0.583$ ,  $\pi_0^+ = 0.047$ ), levels 0–1 contribute 75%.

Figure 2 explains why the tie fraction is small although half of the opinions receive zero votes (Example 5). The tie fraction is a sum of *products* of class shares. The zero-vote class holds 58% of the grounded opinions but only 4.7% of the hallucinations (19 against 1,878 opinions). Hallucinations rarely receive zero votes; they concentrate at twelve votes (38.5%), where almost no grounded opinion is found.

### 7.2 Decomposing the effect of each combination

Because AUROC is an average over pairs (1), it can be decomposed exactly over any partition of the pairs. We divide the 1,314,576 positive–negative pairs into those the committee *separates* (95.4%) and those it *ties* (4.6%), and measure the effect of each combination on each group (Table 2).

Tied pairs contribute exactly  $\tau/2 = 0.0232$  to the committee’s AUROC, as Lemma 2 requires. The weighted combination (“stacking”) extracts information from them, raising their contribution to 0.0249 (+0.0017), but it also reverses 1.33% of the separated pairs, which costs 0.0014, so the net gain is 0.0002. The lexicographic combination reverses no pair and raises the contribution of the tied pairs to 0.0272

<sup>2</sup>Sample, labels and scripts: k1\_auditoria\_planilha.py, k2\_auditoria\_concordancia.py.

**Table 2.** Exact decomposition of AUROC over pair types. “Reversals” counts separated pairs whose order the fused score flips: *harmful* if the committee had them right, *benign* if wrong. One representative fold partition; stability is Table 3.

| Method                 | AUROC         | contribution from |               | reversals among separated pairs |         |        |
|------------------------|---------------|-------------------|---------------|---------------------------------|---------|--------|
|                        |               | separated         | tied          | total                           | harmful | benign |
| committee (vote count) | 0.9260        | 0.9027            | 0.0232        | —                               | —       | —      |
| stacking (logistic)    | 0.9262        | 0.9013            | 0.0249        | <b>1.33%</b>                    | 0.74%   | 0.59%  |
| <b>lexicographic</b>   | <b>0.9300</b> | 0.9027            | <b>0.0272</b> | <b>0.00%</b>                    | 0.00%   | 0.00%  |

(+0.0040). The two methods use the same information; they differ only in whether they may overturn the committee.

### 7.3 Main comparison

**Table 3.** AUROC over twenty independent GroupKFold partitions. The reference, the committee’s vote count (0.9260), does not depend on the folds. “Features” denotes the speaker-incidence features, including NLI; “all” adds the structural features of Section 8. The last column gives the share of partitions in which the combination exceeds the committee.

| Combination                                | median        | min    | max    | > committee |
|--------------------------------------------|---------------|--------|--------|-------------|
| stacking (committee + features)            | 0.9247        | 0.9189 | 0.9276 | 5%          |
| stacking (committee + all)                 | 0.9218        | 0.9173 | 0.9248 | 0%          |
| committee, out-of-fold weights             | 0.9226        | 0.9175 | 0.9245 | 0%          |
| <b>lexicographic (committee, features)</b> | <b>0.9301</b> | 0.9295 | 0.9307 | <b>100%</b> |
| lexicographic (committee, all)             | 0.9302        | 0.9296 | 0.9306 | <b>100%</b> |

Table 3 supports two conclusions. First, stacking is worse than the committee in median, and it is unstable: its result varies by 0.0087 depending only on the fold assignment, a range that includes the committee’s value, so a single run may report either a gain or a loss. Second, the lexicographic combination varies by 0.0012, seven times less, and always exceeds the committee. The paired comparisons on a fixed partition, with hearing-clustered bootstrap, are:

| Comparison                                | $\Delta$ AUROC | 95% CI             |
|-------------------------------------------|----------------|--------------------|
| lexicographic vs. committee (vote count)  | +0.0040        | [+0.0005, +0.0078] |
| lexicographic vs. committee (OOF weights) | +0.0054        | [+0.0005, +0.0105] |
| lexicographic vs. stacking                | +0.0067        | [+0.0015, +0.0125] |

### 7.4 The gain at each level of cost

Table 4 applies the lexicographic combination to committees of 1, 3, 5, 8 and 12 judges. The rule improves every committee size, and its gain decreases as the committee grows, as Equation (3) implies: larger committees tie fewer pairs. The gain is largest at a single judge call. Every one of the twelve judges improves, by +0.046 to +0.102, with all twelve hearing-clustered intervals excluding zero; a typical judge rises from 0.807 to 0.881, and recovers about 60% of the difference between one judge and the full committee. The additional cost is ten calls to a small NLI model; without them the typical gain is still +0.069. With more judges the medians in Table 4 show the same pattern: five judges with the rule (0.923) are above eight without it (0.921). In a paired comparison, the median seven-judge subset with the rule differs from the full committee of twelve by +0.0001 (CI [−0.0046, +0.0049]), so the two are statistically indistinguishable. These comparisons are exploratory. On this dataset, a small committee combined with the rule therefore approaches the accuracy of a much larger one.

**Forecasting the gain on unseen hearings.** Equation (3) is an identity when  $\tau$  and  $\alpha$  are measured on the same data, so agreement with the observed gain on those data confirms nothing. To test whether it forecasts the gain, we split the hearings at random into two halves. On the first half we fit the secondary detector and measure  $\alpha$  with out-of-fold scores; on the second half we compute  $\tau$  from the judges’ votes alone, forecast the gain as  $\tau(\alpha - \frac{1}{2})$ , and compare it with the gain actually observed there, where the secondary detector was never fitted. Table 5 reports the results over 50 random splits, with a random subset of judges in each. The forecast is essentially unbiased at every committee size, and it predicts the sign of the gain in 99% of the 250 cases. The same forecast computed with the global AUROC of the secondary detector instead of  $\alpha$  overestimates the gain by a factor of 1.4 to 3, as Remark 11 anticipates.

**Table 4.** AUROC by number of LLM judge calls per opinion. For  $k$  judges, values are medians over 60 random subsets of  $k$  judges (the single full committee for  $k = 12$ ) on one fold partition. “Alone” is the vote count of the subset; the next two columns add lexicographic tie-breaking with the speaker-incidence features, without and with the ten NLI calls. In the first row, “alone” is the global-cosine baseline and the features are used as the detector. The gain is that of the last column over the first. Values differ from Table 1 in the third decimal because of the fold partition. Script `j9_teto.py`.

| Judge calls | alone  | with tie-breaking |               | gain    |
|-------------|--------|-------------------|---------------|---------|
|             |        | 0 NLI calls       | 10 NLI calls  |         |
| 0           | 0.5676 | 0.7617            | 0.7784        | +0.2108 |
| 1           | 0.8067 | 0.8759            | <b>0.8806</b> | +0.0739 |
| 3           | 0.8917 | 0.9102            | 0.9121        | +0.0204 |
| 5           | 0.9128 | 0.9212            | 0.9230        | +0.0102 |
| 8           | 0.9205 | 0.9255            | 0.9272        | +0.0067 |
| 12          | 0.9260 | 0.9290            | <b>0.9301</b> | +0.0041 |

Since  $\tau$  is known from the votes alone and  $\alpha$  can be estimated on a small development set, the gain of the rule can be estimated before it is deployed.

**Table 5.** Forecasting the gain of the lexicographic combination on unseen hearings: means over 50 random splits of the hearings into two halves. The forecast uses  $\tau$  from the votes of the second half and  $\alpha$  (or the global AUROC) of the secondary detector on the first half. MAE is the mean absolute error of the forecast. Script `j13_previsao.py`.

| Judges | observed gain | forecast with $\alpha$ |        | with global AUROC |        |
|--------|---------------|------------------------|--------|-------------------|--------|
|        |               | mean                   | MAE    | mean              | MAE    |
| 1      | +0.0721       | +0.0706                | 0.0108 | +0.1000           | 0.0279 |
| 3      | +0.0284       | +0.0270                | 0.0083 | +0.0490           | 0.0206 |
| 5      | +0.0152       | +0.0148                | 0.0062 | +0.0310           | 0.0159 |
| 8      | +0.0075       | +0.0073                | 0.0044 | +0.0185           | 0.0111 |
| 12     | +0.0042       | +0.0041                | 0.0032 | +0.0130           | 0.0088 |

**The gain expressed in judge calls.** Over twenty fold partitions and sixty random subsets of judges per subset size, eight judges combined lexicographically with the features reach 0.9272, against 0.9260 for the full twelve-judge committee. The difference, +0.0008 (CI  $[-0.0031, +0.0048]$ ), is non-inferior at a margin of 0.0040, the size of the effect claimed in this paper. The combination matches or exceeds the full committee in 73% of the eight-judge subsets and in 97% of the nine-judge subsets. In this exploratory analysis, eight or nine judges combined with the features therefore match the full committee in most random subsets, which would save three to four judge calls per opinion. Two limitations apply. The choice of judges matters more than the fold assignment (standard deviation 0.0025 across subsets against 0.0004 across folds, at seven judges), so a correlated choice, such as four models with the same prompt, performs worse than a random one. With seven judges or fewer, the shortfall exceeds the margin. This analysis is exploratory.

**Individual judges.** Applied to each judge separately, the lexicographic combination outperforms stacking for all twelve judges, six of them with clustered CIs excluding zero, and the largest gains go to the judges with the most ties. The headline single-call figures of this paper are those of a typical judge (Table 4); the best judge is a variant selected after the fact. For that judge (p2\_gpt4omini, 0.8468), the combination reaches 0.8928, against 0.8877 with stacking; adding the structural features of Section 8 gives 0.8942.

**Where the combination is useful.** Both  $\tau$  and  $\alpha$  depend on how coarse the primary detector is. The best single judge (AUROC 0.8468) ties  $\tau = 26.1\%$  of the pairs, a maximum gain of 0.1307, which is 5.6 times that of the twelve-judge committee. On these ties the speaker-incidence features reach  $\alpha = 0.676$ , against 0.5856 on the committee’s ties. Equation (3) accounts exactly for the observed gains, +0.0460 (0.8468  $\rightarrow$  0.8928) and +0.0040 (0.9260  $\rightarrow$  0.9300). The gain is eleven times larger with one judge than with twelve, and the maximum gain  $\tau/2$ , which is known from the judges’ votes alone, already indicates this.

## 7.5 How much of the possible gain is obtained, and why not more

Equation (3) expresses the observed gain as a property of the secondary detector. On the representative partition, the speaker-incidence features reach  $\alpha = 0.5856$  on the committee’s ties (median 0.589 over

twenty partitions), so the combination recovers  $2\alpha - 1 \approx 17\%$  of the maximum gain  $\tau/2$ . The value  $\tau(\alpha - \frac{1}{2})$  equals the observed +0.0040; we verified the identity numerically to six decimal places. Table 6 shows how this value arises.

**Table 6.** AUROC of the speaker-incidence features on each group of pairs, by what the committee does with the pair; medians over twenty fold partitions. The middle row is the maximum gain of Proposition 10(iii). The last row limits everything else: where the committee is wrong, the features are wrong as well.

| the committee...          | % of pairs | contributes   | features' AUROC |
|---------------------------|------------|---------------|-----------------|
| separates, correctly      | 90.3%      | 0.9027        | 0.809           |
| <b>ties</b>               | 4.6%       | <b>0.0232</b> | <b>0.589</b>    |
| separates, <b>wrongly</b> | 5.1%       | 0.0000        | <b>0.473</b>    |

The maximum gain is highly concentrated (Figure 2). Three quarters of it depends on the 37 hallucinations (19 + 18) that received zero or one vote. These are, by construction, the hardest cases in the dataset, since eleven or twelve judges did not flag them, and only 19 of them lie in the zero-vote class, against 1,878 grounded opinions. On these cases the features reach  $\alpha = 0.56$ ; where they perform well (0.87 at twelve votes), only 2.3% of the maximum gain is available. The structural features of Section 8 change  $\alpha$  from 0.5856 to 0.5852.

**Limits of the remaining gap.** The features reach 0.809 on the pairs the committee orders correctly, 0.589 on the pairs it ties, and 0.473, below chance, on the pairs it orders incorrectly; the last value never exceeds 0.482 across twenty partitions. The features therefore agree with the committee precisely where the committee is wrong. This limits both remaining sources of gain. Ordering the ties can recover only what  $\alpha = 0.589$  allows, and no rule that overrules the committee when the features are confident can recover the 0.0508 of AUROC lost to the committee's incorrect orderings, since disagreement with the committee is not a sign of committee error. Table 2 shows the same effect: only 44% of the pairs that stacking reversed were in the wrong order.

We tested this conclusion directly. Six pre-specified attempts to raise  $\alpha$  (fitting the secondary detector within each committee class, adding the vote count as a feature, interactions with the vote count, interactions with four coarse strata, weighting by the mass of tied pairs, and a pairwise ranking loss on the tied pairs only) all *lower* it, from a median of 0.589 to between 0.489 and 0.520, in all twenty fold partitions. The limit is not due to the model class either: gradient-boosted trees on the votes and all features reach 0.9042, below the committee alone. One of these failures yields a practical rule: *the secondary detector should not receive the primary detector's output as input*. With the vote count as a feature, the secondary detector's global AUROC rises to 0.9260, because it relearns the committee, while its  $\alpha$  falls to 0.514. The norm of its speaker-incidence coefficients halves ( $0.921 \rightarrow 0.470$ ) and their direction rotates (cosine 0.889). Since the vote count is constant within a tied class, only this rotated and weaker direction remains to order the ties.

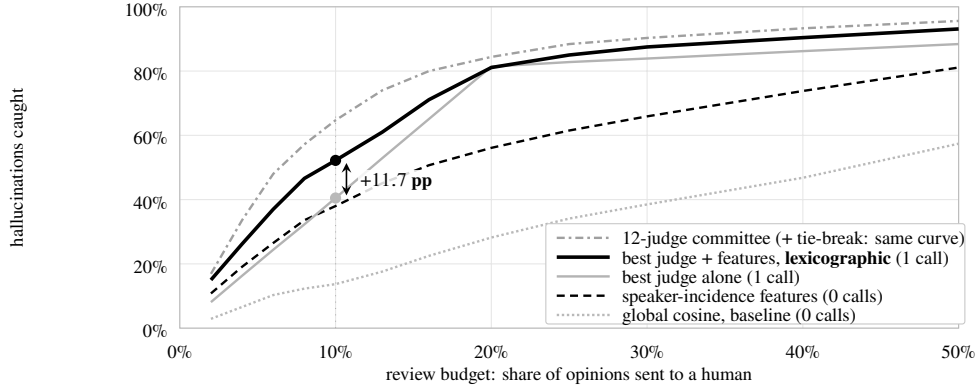
Finally, the combinations  $f_\gamma$  of Remark 12, which allow the secondary detector to move an item across up to  $\gamma$  classes of the committee, do not help with the actual features:  $\gamma = 2$  exceeds the lexicographic combination by only +0.0005 (CI  $[-0.0004, +0.0013]$ ) and exceeds its median in 85% of the twenty partitions, while giving up the guarantee of Proposition 10(i). The mechanism is not the limitation: with a perfect secondary detector,  $f_\gamma$  ranges from 0.9492 to 0.9975 as  $\gamma$  grows. To estimate how good a new feature would need to be, we replaced the secondary detector by the true label, flipped at random for a fraction  $\rho$  of the opinions, and for each  $\rho$  between 0 and 0.5 searched for a value of  $\gamma$  at which  $f_\gamma$  outperforms the lexicographic combination. Overruling the committee helps only while the detector's AUROC on the committee's incorrectly ordered pairs exceeds approximately 0.58.<sup>3</sup> The speaker-incidence features reach 0.473 on those pairs.

## 7.6 Effect on a human review queue

AUROC concerns pairs of opinions, whereas a human reviewer works through a queue, from the most to the least suspect opinion, until the review budget is exhausted. Figure 3 reports, for each detector, the share of the 408 hallucinations found when the top  $k\%$  of opinions are sent to a human reviewer.

At the inexpensive levels, the operational gain is large and agrees with the theory. When one opinion in ten is reviewed, the global-cosine baseline finds 13.7% of the hallucinations, the speaker-incidence features 38.0% at the same zero cost, and a single judge combined with them 52.2%, against 40.5% for

<sup>3</sup>Script `j10_portao.py`; the value is interpolated between the tested levels of  $\rho$ .



**Figure 3.** Share of hallucinations found as a function of the review budget. The curve of the best judge alone is a straight line up to 20%: a binary verdict does not order the opinions within each of its two classes, so the queue is in random order there. Ordering these ties with the speaker-incidence features adds 11.7 percentage points at a 10% budget with one judge call; with twelve judges the difference is not measurable.

the same judge alone. This single-judge gap follows from Equation (3) with  $\tau = 26.1\%$ : a binary judge ties most pairs, so most of its queue is unordered.

With twelve judges, however, the gain in the queue is not measurable. We predicted in advance that the lexicographic queue would outperform the committee’s at every review budget. It does not: the differences are  $-0.2$ ,  $+0.3$  and  $+0.6$  percentage points at budgets of 5%, 10% and 20%, and every hearing-clustered interval includes zero. The committee’s own queue is ambiguous, since its thirteen levels leave the achievable recall within an interval of 1.7 to 4.7 points, so two reviewers with the same committee and budget may not review the same opinions. This ambiguity is small, however, and a gain of  $\tau/2 = 0.0232$  in AUROC is too small and too spread out to change a review budget. At twelve judges the result is therefore a ranking improvement, not a workload improvement; at one judge it is both.

## 7.7 Controls

**Random tie-breaking.** Ordering the ties at random has the same expected AUROC as not ordering them, but a non-negligible variance: over 200 independent draws the mean is 0.9259 (standard deviation 0.0020, maximum 0.9308). A single random draw is therefore not an adequate control. Against this null distribution, the lexicographic result of 0.9300 has  $p = 0.015$ .

**A weak secondary detector, and the order of the combination.** Using the global cosine as the secondary detector ( $\alpha = 0.5409$ ) gives 0.9279, a difference of  $+0.0021$   $[-0.0017, +0.0057]$  that is not significant, as is consistent with its low  $\alpha$  under Equation (3). The lexicographic combination is also asymmetric by design: the committee comes first because it is the more reliable detector, and reversing the order would lose the guarantee of Proposition 10(i).

## 8 Negative Results

This section reports the pre-specified predictions that the data contradicted. Two others appear in the results: conditioned entailment does not outperform conditioned cosine similarity (Section 6), and at twelve judges the combination does not improve the review queue (Section 7.6).

**Structural features add nothing to the combination.** We built five features from the relation between opinions and speakers: its homology [3], a Hodge decomposition [7] of attribution discrepancies, duplication between opinions, and two constraints on the order and exclusivity of the support. Individually, each is either uninformative (the relation has no cycles in any hearing; the constraints are already satisfied by 87 to 93% of the opinions) or redundant with the speaker-incidence features. Together they raise the zero-cost detector from 0.7611 to 0.7718 (paired  $+0.0107$ , CI  $[+0.0012, +0.0205]$ ), but they leave the combination with the committee unchanged (0.9302 median in both cases). The gain reported in this paper is due to the combination rule, not to these features.

**Attempts to improve the secondary detector on the ties fail.** Fitting the secondary detector separately within each committee class gives 0.9237, below a single global model, because the class that holds most of the maximum gain contains only 19 positives. This and the five other pre-specified attempts described in Section 7.5 all lower  $\alpha$ .

## 9 Threats to Validity

**Labels.** The hallucination labels are those released with PublicHearingBR [5], which we use as given; our conclusions are conditional on their quality, and no inter-annotator agreement is available for them. The audit of Section 6.3 characterises the positives but does not replace the labels.

**Binary judges.** The judges’ votes are binary, so the maximum gain  $\tau/2$  would shrink if the judges produced calibrated probabilities. Our result on combination is conditional on this format.

**Speaker recovery.** Speakers are matched by name for 85.7% of the opinions, and matching errors are not necessarily conservative: excluding the hearings with deaf participants, whose turns read “(*Manifestação em LIBRAS.*)” while the content appears in the interpreter’s voice (1.1% of the opinions), lowers the main signal from 0.7455 to 0.7314. All entailment probabilities come from a single multilingual NLI model.

**Scope of the claims.** The gain over the full committee is small (+0.0040, clustered CI lower bound +0.0005) and rests on its consistency: it holds in all twenty fold partitions and for all twelve individual judges ( $p = 0.015$  against random tie-breaking). The share of partitions measures stability on a fixed dataset, not replication. The advantage over stacking (+0.0067) is more robust. The confirmatory comparisons are the three of Section 7.3 and the six attempts to raise  $\alpha$ ; the per-judge results, Table 4 (one fold partition, medians over random subsets of judges), the decomposition of Section 7.5 and the audit are exploratory.

## 10 Conclusion

Verification scores that pool sentence-level similarity or entailment over a whole transcript cannot, in principle, tell whether an opinion was expressed by the person to whom it is attributed. On PublicHearingBR, conditioning on the attributed speaker’s turns raises the AUROC of an inexpensive detector from 0.57 to 0.76 without any model call, and improves each of the twelve LLM judges. An audit of 50 positives indicates that about a quarter of them are misattributions of this kind.

Adding such a detector to a committee of judges requires care. Fitting a single model on both reverses some of the committee’s decisions, and on this dataset it cancels the benefit. Ranking by the committee and using the inexpensive detector only to order its ties never reverses a decision, and its gain is given exactly by the tie fraction and by the detector’s AUROC on the tied pairs. The tie fraction, known from the judges’ votes alone, bounds the gain and indicates where the combination is useful. The main practical consequence is a large reduction in cost. With a single judge call, the rule raises every judge’s AUROC, a typical judge from 0.81 to 0.88, which recovers about 60% of the difference to the full committee; when a reviewer checks the 10% most suspect opinions, the share of hallucinations found rises from 40.5% to 52.2%. In an exploratory analysis, seven judges with the rule are statistically indistinguishable from the full committee of twelve, and the full committee itself improves from 0.926 to 0.930. Because the gain decomposes into two quantities that can be measured in advance, it can also be forecast on new data, which we verified on held-out hearings. The remaining gap cannot be closed with these features, since they perform below chance on the pairs that the committee orders incorrectly.

Several predictions failed, and we report them alongside the results that held. The proofs use only the rankings induced by the scores, so the results apply to any primary and secondary detector and can be checked on a new deployment by counting pairs.

**Reproducibility.** All numbers in this paper are produced by the scripts in the `codigo/` directory of the public repository (<https://github.com/caua-sathler/IdeiasEmRede-TopoIA>), together with the prediction files written before each experiment. The pipelines use cached intermediate results and fixed random seeds, and each runs in minutes on a single laptop with 8 GB of memory. We thank Instituto Kunumi for the *Ideias em Rede* challenge, the authors of PublicHearingBR for the corpus and the judge annotations, and the DCC/UFMG course infrastructure.

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